Mayank Raj Gupta

Full-Stack Developer | BCA Scholar @ Christ University, Bangalore
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SUMMARY

Full-Stack & Software Developer studying at Christ University with practical experience in Java, Python, React, Next.js, and TypeScript. Familiar with core Data Structures & Algorithms, modern UI engineering, and computer vision workflows. Seeking a Software Engineering Internship to gain hands-on industry experience, learn from senior developers, and contribute to team projects.

EDUCATION

- Bachelor of Computer Applications (BCA), Christ (Deemed to be University), Bangalore (2025 – 2029) – 3.33/4.0 CGPA
  Coursework: DSA, OOP (Java), DBMS, Web Architecture
- Class XII (CBSE), Rajkamal Saraswati Vidya Mandir, Dhanbad, Jharkhand (2023)

EXPERIENCE

Web Development Intern – ShadowFox *[May 2026]*
Virtual Internship • Remote

- Completed an intensive web development internship building responsive UI components and frontend modules.
- Developed modular web pages following modern responsive design principles, earning a Certificate of Completion (SF05VIP397).

AI & Computer Vision Research Project Lead – Christ University *[2026 – Present]*
School of Sciences / CS Department • Bangalore

- Led confidential research fine-tuning YOLOv11 deep learning architectures for real-time vehicle and license plate localization.
- Engineered image preprocessing pipelines in OpenCV using adaptive thresholding and CLAHE to achieve 94%+ detection accuracy.

PROJECTS

Wholesale & Retail E-Commerce Platform (Roshan Enterprises) *[December 2025]*
Demo: e-commerce-roshan-enterprises-dhn.web.app | GitHub: github.com/M20A03/E-commerce-Online-Shopping-for-Retail-and-Wholesale
React, Vite, Firebase

- Architected a production-ready B2B/B2C marketplace supporting dual wholesale and retail checkout flows with role-based auth.
- Implemented list virtualization via *react-window* for 60fps scrolling across 50+ catalog items with zero cumulative layout shift.
- Enforced WCAG 2.1 AA accessibility compliance and responsive mobile-first safe area styling across viewports.

DSA Search Algorithm Visualizer & Space Explorer *[February 2026]*
Demo: linear-and-binary-search.vercel.app | GitHub: github.com/M20A03/Linear-and-Bineary-Search-Simulation
Vanilla JavaScript, HTML5 Canvas, WebAudio API

- Engineered an interactive visual simulation platform for Linear and Binary Search with dynamic execution pointer highlights.
- Integrated 'Star-Command AI' conversational assistant to explain Big-O time and space complexity to learners.
- Synthesized real-time frequency-modulated audio feedback using the WebAudio API for each comparison step.

HACKATHONS & CERTIFICATIONS

- **Hackathons –** PromptWars (Google Devs x Christ Univ), InHack '26 & HackNova, CODEX '26 (Don Bosco), YUGASTR '26
- **Certifications –** SPARK Research Program (Christ Univ), ShadowFox Web Dev Intern, AI Prompt Engineering (MSFT/Reliance), Infosys C

TECHNICAL & PROFESSIONAL SKILLS

- **Programming Languages –** Java, Python (Pandas, NumPy, Tkinter, CSV), C, C++, JavaScript (ES6+), TypeScript, SQL (PostgreSQL, MySQL)
- **AI & Machine Learning –** PyTorch, Whisper ASR, Mistral-7B, YOLOv11, OpenCV, Deep Learning, Computer Vision
- **Web & Backend –** Android App Dev (React Native, Expo), React 19, Next.js 15, FastAPI, Node.js, Express.js, Tailwind CSS
- **Databases & Cloud –** PostgreSQL, MySQL, Firebase, Supabase, Railway, AWS (S3, EC2), Vercel
- **Tools & Competencies –** Git, GitHub, Postman, Linux/Bash, VS Code, Eclipse IDE, DSA, Problem Solving, Agile Teamwork
- **Spoken Languages –** English (Professional / Fluent), Hindi (Native)